

# Ton Nu Thanh Thao

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## OBJECTIVE

As a senior at Fulbright studying Human-Centered Engineering, I am passionate about both teaching children and developing educational products. In particular, I am interested in using my technical knowledge of electronics, programming, and system design to create educational engineering products for children.

## SKILLS

*Design:* Circuit Design, Altium, Autodesk Fusion 360, Tinkercad, Canva

*Programming:* Python, C++, Jupyter Notebook, MATLAB

*Technical:* Arduino IDE, Ubuntu, VMware, Robot Operating System (ROS), Circuit Soldering & Debugging, MS Office

*Soft Skills:* Team Leadership, Problem-Solving, Time Management, Event Management

*Language:* English (Fluent), Vietnamese (Native), Chinese (Beginner)

## EDUCATION

**Fulbright University Vietnam** | Ho Chi Minh City, Vietnam

August 2023 - Present

*Bachelor of Engineering. Major:* Human-Centered Engineering

GPA: 3.76/4.0

*Minor:* Applied-Mathematics & Computer Science

## WORK EXPERIENCE

**Bosch Automotive R&D Center in Vietnam, Technical Documentation Intern**

August 2025 - February 2026

- Aided in analyzing 10,000 technical legal requirements and texts of the Indian and Brazilian markets, relevant to product development for Bosch mobility products in two technical topic areas (Safety and Security)
- Checked the new version and identified quality issues of 20+ relevant laws/regulations and standards of Brazil and Korea from the existing database
- Developed user-friendly resources (Docupedia and a one-pager) that streamlined the team processes

**VUIHOC.vn, English Tutor**

August 2025 - Present

- Taught English online in 1:1 and 1:4 classes for middle and primary school students via ClassIn software
- Conducted 16 classes every week, and each class lasted 45 minutes
- Provided brief feedback on student learning progress after each lesson

## COURSE PROJECTS

**Analog-only solar panel auto charging and auto tracing to light source**

January 2026 – May 2026

- Designed an analog-only solar-tracking system with 2 distinct LDR sensors and LM358 comparators with hysteresis
- Simulated a comparator hysteresis system prevent jiggling by using Altium to ensure the device runs accurately
- Developed a self-sustainable solar charging system in cloudy days

**AURA BAND: Real-time voice-activated safety device for the visually impaired**

January 2026 – May 2026

- Implemented real-time DSP pipeline on ESP32 with INMP441 I2S audio, IIR filtering, and on-device ML (Edge Impulse) for voice command recognition
- Architected a dual-core FreeRTOS task scheduler, separating audio capture and inference
- Built a prototype with NeoPixel LED and buzzer outputs corresponding to 6 voice commands

**Line Follower Obstacle Avoiding Robot & Navigation and Bottle Grabbing Robot**

August 2024 – December 2024

- Built a real-time Line Follower Obstacle Avoiding Robot with ROS2 and ultrasonic sensors
- Created a Bottle Grabbing Robot with A\* path planning and LiDAR-based mapping
- Led a three-member team in debugging issues from physical connections to programming for correct motion

### Mini Boat and Mini Submarine

September 2024 – December 2024

- Programmed a Mini Boat and a Mini Submarine using an ESP32 microcontroller and IMU sensors, with Wi-Fi-based remote control and real-time stability adjustment for enhanced navigation
- Calculated and designed the prow using Autodesk Fusion 360 to improve hydrodynamic stability and reduce resistance for faster movement
- Guided 3 members in troubleshooting of soldering processes and ESP32 communication errors

### MINDSTRESS Device

April 2024 – June 2024

- Developed a MINDSTRESS device using the Design Thinking Process to detect the emotional status based on the heart rate and suggest meditation music for relaxation
- Designed a prototype of the device using an Arduino Uno and MAX30102 sensors
- Illustrated the product's circuit and 3D design using Tinkercad and Autodesk Fusion 360

### Tank Toy

February 2024 – April 2024

- Used Autodesk Fusion 360 to create a wooden chassis model based on a provided template
- Soldered and assembled electronic components, including motor drivers, MP3 modules, and power circuits
- Distributed completed Tank Toy to children in Go Vap's SOS Village

## RESEARCH EXPERIENCE

### Capstone Project Research about [SOLIX Socially Aware Robot](#), *Research Assistant*

January 2025 – October 2025

- Investigated and compared Socially Aware Motion Planning systems to improve the method with group interaction detection and human activity recognition
- Contributed to the design and development of a new approach to enhance responsiveness, stability, and smoothness in real-world applications
- Collaborated with system integration and product development to combine computational modeling and hardware design principles to deliver enhanced real-world robotic applications

### Independent Study about **Swarm Robotics System**, *Robotics Developer*

March 2025 – August 2025

- Co-designed and discovered new algorithms (ROS2 Navigations) for SLAM, trash classification, and detection with Color Detection
- Tested with a swarm robotics system to validate system performance under dynamic environmental situations
- Researched and analyzed multi-robot control frameworks to create a new optimized method with efficient task allocation and real-time path planning

## VOLUNTEER

### The Route to Knowledge - **SUGAR Vietnam**, *Project Leader & Volunteer Tutor*

March 2022 – October 2022

- Provided 1:1 Mathematics and Literature tutorship for five students in 36 hours
- Administered and evaluated the teaching process of a project of 50+ members
- Organized Orientation Day and Training Day about Children's Understanding and Teaching skills for 30+ volunteers

### 17th RED FLAMBOYANT VOLUNTEER CAMPAIGN 2022, *Team Leader*

June 2022 – August 2022

- Organized trips to "The Future Disability School" (with 30+ participants) and Dieu Giac Orphanage with 60+ participants to give kids school stationery and create playgrounds
- Delegated event organizing work to personnel of 40+ members to support 5+ shelters
- Planned two performances for the music show Light Up Dream attracting 400+ audiences to raise \$500 for homeless children and run seven charitable activities in Ho Chi Minh City

## PUBLICATIONS

- Nguyen, T. L., Nguyen, D. P., Nu, T. T. T., Le, Q., Tran, T. H., & Phung, M. D. (2025). [Real-time recognition of human interactions from a single RGB-D camera for socially-aware robot navigation](#). *Intelligent Service Robotics*, 18(6), 1369-1380.